

torch.fake_quantize_per_channel_affine#

https://pytorch.org/docs/stable/generated/torch.fake_quantize_per_channel_affine#

Description:

Returns a new tensor with the data in input fake quantized per channel using scale, zero_point, quant_min and quant_max, across the channel specified by axis. input (Tensor) – the input value(s), in torch.float32 scale (Tensor) – quantization scale, per channel in torch.float32 zero_point (Tensor) – quantization zero_point, per channel in torch.int32 or torch.half or torch.float32 axis (int32) – channel axis

Function Signature

```
torch.fake_quantize_per_channel_affine(input, scale,  
zero_point, axis, quant_min, quant_max) → Tensor#
```

Parameters

Returns

Return type

Returns:

Tensor

Code Examples

```
>>> x = torch.randn(2, 2, 2)
>>> x
tensor([[[[-0.2525, -0.0466],
          [ 0.3491, -0.2168]],

         [[-0.5906,  1.6258],
          [ 0.6444, -0.0542]]]])
>>> scales = (torch.randn(2) + 1) * 0.05
>>> scales
tensor([0.0475, 0.0486])
>>> zero_points = torch.zeros(2).to(torch.int32)
>>> zero_points
tensor([0, 0])
>>> torch.fake_quantize_per_channel_affine(x, scales,
zero_points, 1, 0, 255)
tensor([[[[0.0000, 0.0000],
          [0.3405, 0.0000]],

         [[0.0000, 1.6134],
          [0.6323, 0.0000]]]])
```

Detailed Documentation

Documentation

Returns a new tensor with the data in input fake quantized per channel using scale, zero_point, quant_min and quant_max, across the channel specified by axis.

input (Tensor) – the input value(s), in torch.float32

scale (Tensor) – quantization scale, per channel in torch.float32

zero_point (Tensor) – quantization zero_point, per channel in torch.int32 or torch.half or torch.float32

axis (int32) – channel axis

quant_min (int64) – lower bound of the quantized domain

quant_max (int64) – upper bound of the quantized domain

A newly fake_quantized per channel torch.float32 tensor